STEIN VS FISHER SCORE

14

$$U_\beta = U_0 + \beta \Delta U$$

$$\Delta U = U_1 - U_0 = -\log w$$

- Encodes the direction mass changes between distributions

$$\pi_{\beta'} \approx \pi_\beta + \Delta\beta \dot{\pi}_\beta$$

- Encodes the geometry of the normalising constant and modes

Fisher Score

Stein Score

$$\nabla \log \pi_{\beta}(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}\beta} \log \pi_\beta(x)$$

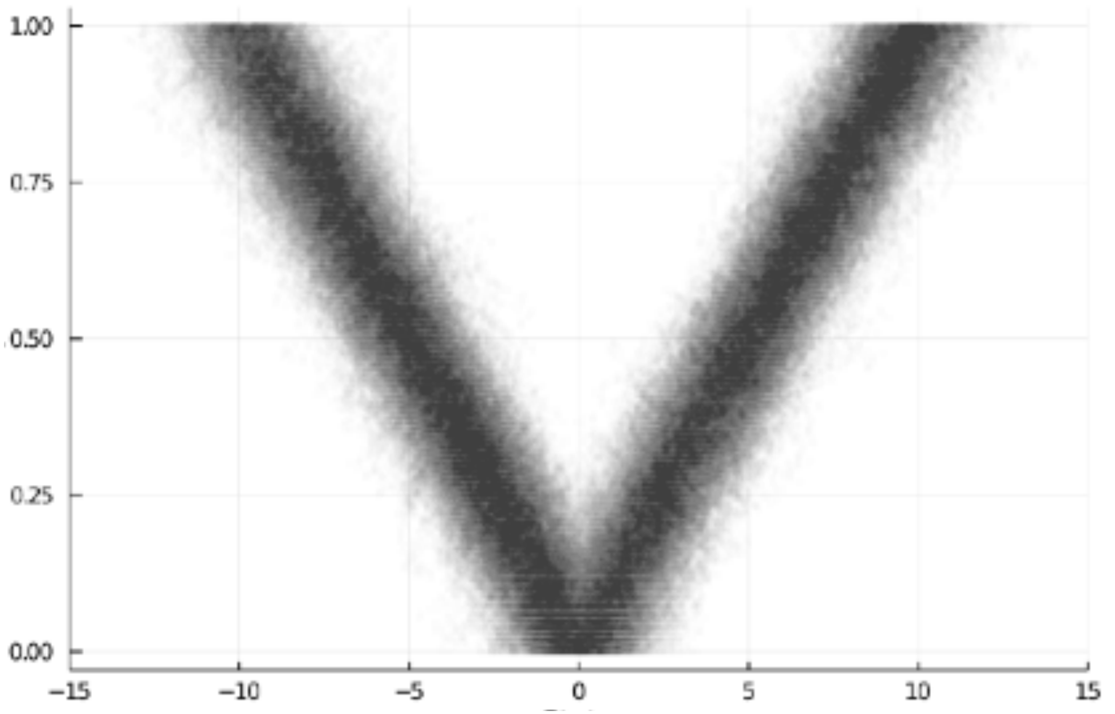
- Encodes the geometry of the modes

$$X_{t'} \approx X_t + \Delta t \, \nabla \log \pi_\beta(X_t)$$

- Encodes the geometry of the modes

Controls for annealing

- Controls local inference


1.00
0.75
0.50
0.25
0.00
−15
−10
−5
0
5
10
15

STEIN VS FISHER SCORE

Fisher Score

$$\frac{\mathrm{d}}{\mathrm{d}\beta} \log \pi_\beta(x)$$

- Encodes the direction mass changes between distributions

$$\pi_{\beta'} \approx \pi_\beta + \Delta\beta\dot{\pi}_\beta$$

- Encodes the geometry of the normalising constant and modes
- Controls for annealing

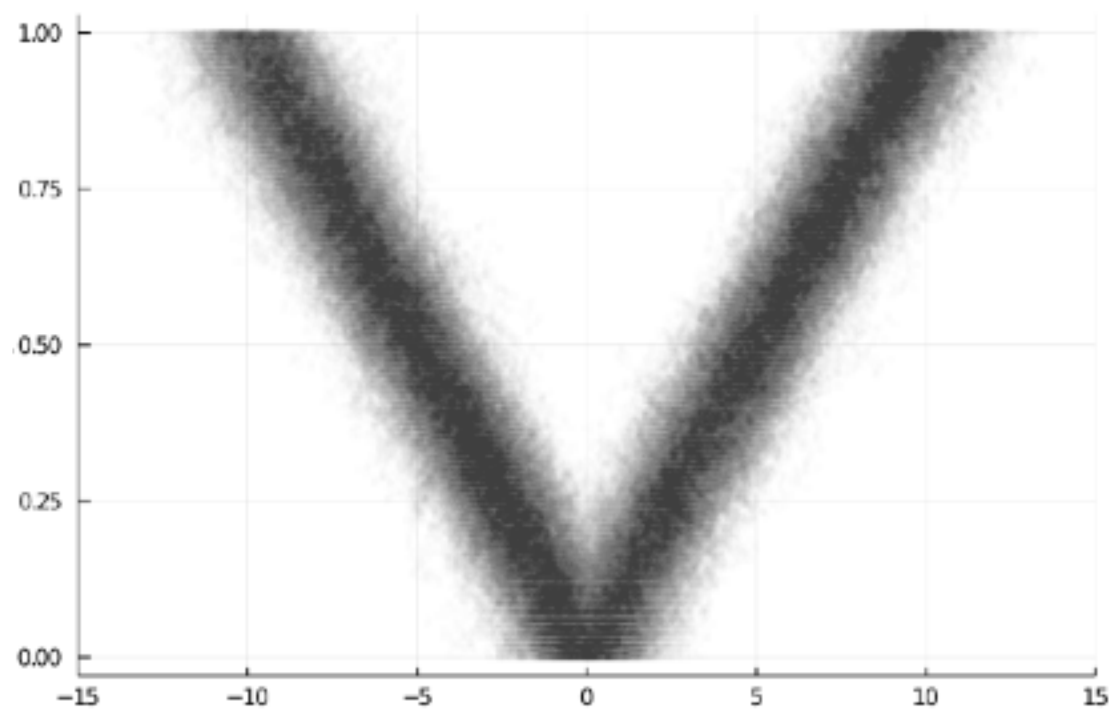
Stein Score

$$\nabla \log \pi_\beta(x)$$

- Encodes the geometry of the modes

$$X_{t'} \approx X_t + \Delta t \nabla \log \pi_\beta(X_t)$$

- Encodes the geometry of the modes
- Controls local inference



LINEAR PATH AS AN EXPONENTIAL FAMILY